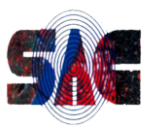


Chandrayaan-3 Navigational Camera (NavCam) PDS4 Data Products and Archive Software Interface Specification



**Planetary and Space Science Data Processing Division
Planetary and Meteorology Data Processing Group
Signal & Image Processing Area
Space Applications Centre, ISRO
Ahmedabad – 380 015**

This page is intentionally left blank

	अंतरिक्ष उपयोग केन्द्र I Space Applications Centre अहमदाबाद I Ahmedabad	
DOCUMENT CONTROL DATA SHEET		
1	Title:	Chandrayaan-3 Navigational Camera (NavCam) PDS4 Data Products and Archive Software Interface Specification
2	Author (s) / Project Team:	• Data Processing • Payload • Application
3	Division:	SAC/SIPA/PMPG/PSPD
4	Affiliation of Author(s) other than SAC:	
5	Collation (No. of Pages, maps, etc.):	61
6	Abstract:	This document provides the Data Products and Archive Software Interface Specifications of NavCam instrument of Chandrayaan-3 Rover
7	Keywords:	• Archive • PDS4 • ISDA
8	Date of Submission:	18-12-2023
9	Security & Distribution Status	Unrestricted
11	Report No.:	ISRO-SAC-PR-CHANDRAYAAN-3-2023-011

Chandrayaan-3

Rover Navigational Camera (NavCam)

PDS4 Data Archive

Software Interface Specifications

Document Change History

*A: Addition; M: Modification; D: Deletion

Version No.	Date	Section No.	A/M/D *	Description of change
V 1.0	December, 2023	All	A	First Release Version
V 2.0	July 3, 2024	1.5	A	Added reference for CAHV algorithm
		1.6	A	Added CAHV acronym under table 1
		3.2	A	Added new section on the exposure settings for NavCam payload
		6.4.2	A	Added CAHV information under the section and added CAHV files under table 17 calibration contents products
		Appendix-H: CAHV Camera Model	A	Added details about the CAHV models
		Appendix-I: Rover Mobility Summary	A	Added the table shows the Rover movement as per the commands sequence operated by mission team
		Appendix-J: Rover NavCam Exposure Settings	A	Added the table shows product wise exposure settings configured during the whole mission.
		Appendix-K: Rover NavCam Gain Settings	A	Added the table shows product wise gain settings configured during the whole mission.

Acknowledgements

Authors express their sincere gratitude to Shri. Nilesh M Desai, Director, Space Applications Centre for providing the opportunity to carry out for development of Chandrayaan-3 Rover Navigational Camera (NavCam) Data Products Software Interface Specifications & Archive. Special thanks to Shri. Debajyoti Dhar, Deputy Director, SIPA and for his constant motivation and guidance. Thanks are also due to Dr. Nitant Dube, Group Director, PMPG for his guidance and support during this work. Last but not the least we would like to extend our thanks to PDS4 working group members and All the PI's of Chandrayaan-3 for their support.

Technical Content Approvals

		Name & Function
	Reviewed & Approved by	Amitabh DPD (DP) Chandrayaan-3

Content

Document Change History	4
<i>Acknowledgements</i>	5
Technical Content Approvals.....	6
Content.....	7
1. Introduction	10
1.1 Purpose.....	10
1.2 Scope	10
1.3 Data Products and Archive SIS Content Overview	10
1.4 Audience.....	10
1.5 Applicable Documents	11
1.6 Abbreviations Acronyms, and Description	13
1.7 Glossary	14
2. Mission Overview.....	16
2.1 Mission Objective	16
2.2 Payload Objective	17
3. Instrument Overview	18
3.1 Specifications	18
3.2 Exposure Settings.....	19
4. Data Overview.....	20
4.1 Data Processing Levels	20
4.2 Data Products	21
4.2.1 Raw Data Products	21
4.2.2 Calibrated Data Products	23
4.3 Data Products Bundle.....	24
4.4 Data Flow.....	24
5. Archive Generation	25
5.1 Data Processing and Production Software	25
5.2 Data Validation	25
5.3 Data Transfer Methods and Delivery Schedule	25
5.4 Data Product and Archive Volume Size Estimates.....	26
5.5 Backups and duplicates	26
6. Archive Organization and File Naming Conventions.....	27
6.1 Namespace Registration.....	27
6.2 Logical Identifiers	27

6.3	Mission Archive Structure at Bundle Level.....	28
6.4	Instruments Collection	29
6.4.1	Root directory content	30
6.4.2	Calibration directory content	30
6.4.3	Context directory content	31
6.4.4	Document directory content.....	32
6.4.5	Data directory content	32
6.4.6	Xml_schema directory content	33
6.5	File Naming Conventions & Formats	34
7.	Archive Products formats.....	36
7.1	Data File Formats.....	36
7.2	Document Product File Formats.....	36
7.3	PDS Labels	36
7.3.1	XML Documents	36
7.4	Delivery Package.....	37
	Appendix A Support staff and cognizant Persons	38
	Appendix B Sample Bundle Product Label	39
	Appendix C Sample Collection Product Label	41
	Appendix D Sample Data Product Labels	44
	Appendix E Local Data Dictionary	47
	Appendix F Data Disclaimer	49
	Appendix-G: Raw to Calibrated Data Conversion	50
	Appendix-H: CAHV Camera Model.....	51
	Appendix-I: Rover Mobility Summary	52
	Appendix-J: Rover NavCam Exposure Settings	55
	Appendix-K: Rover NavCam Gain Settings	59

List of Figures

Figure 1 Mission Profile.....	17
Figure 2 NavCam Left and Right on Rover	18
Figure 3 NavCam Raw Data Archive	22
Figure 4 NavCam Calibrated Data Archive	23
Figure 5 Chandrayaan-3 Mission Level Archive Organization	29
Figure 6 Chandrayaan-3 nav Instrument wise collection structure	29

List of Tables

Table 1 Acronym, Abbreviations and Description	13
Table 2 PDS4 Glossary	14
Table 3 Specification of NavCam	18
Table 4 Chandrayaan-3 NavCam Archive Schema and Schematron	20
Table 5 Chandrayaan-3 NavCam Data Processing Levels	20
Table 6 Chandrayaan-3 NavCam Data Products Definitions	21
Table 7 NavCam Raw Data Products	22
Table 8 NavCam Calibrated Data Products	23
Table 9 NavCam Data Products Bundles	24
Table 10 Data Release Schedule Plan.....	26
Table 11 ISDA Namespace for PDS4.....	27
Table 12 LID for Chandrayaan-3 mission.....	27
Table 13 LID for Chandrayaan-3 mission instruments	28
Table 14 Mission Attributes defined at structure level.....	28
Table 15 Collection product types	30
Table 16 Root directory content products	30
Table 17 Calibration directory content products	31
Table 18 Context directory content products	32
Table 19 Document directory content products.....	32
Table 20 Xml_schema directory content products	33
Table 21 File Naming Conventions	34
Table 22 PDS4 File Names and Data Formats	35
Table 23 NavCam PDS4 Data Products.....	36
Table 24 Chandrayaan-3 Team staff	38
Table 25 ISDA Local Data Dictionary Class and Attributes List.....	47

1. Introduction

1.1 Purpose

The Data Products and Archive software interface specifications describes the data products, format and content of the Navigational Camera (NavCam) Planetary Data System (PDS) data archive bundles in which data products with associated meta data, documentation, and supporting materials are stored. This document is intended to provide enough information to enable users to read and understand the NavCam data products stored in PDS.

1.2 Scope

The specifications in this document apply to all NavCam data products submitted for archival to the ISRO Science Data Archive (ISDA), for all the mission phases of the Chandrayaan-3.

1.3 Data Products and Archive SIS Content Overview

Section 2 of this document describes the Mission overview while Section 3 provides the characteristics of the Navigational Camera (NavCam). Section 4 gives an overview of data organization and data flow while Section 5 describes data archive generation, delivery, and validation. Section 6 describes the archive structure and archive production responsibilities. Section 7 describes the file formats used in the archive, including the data product record structures. Individuals involved with generating the archive volumes are listed in Appendix-A While Appendix-B contains sample bundle label, Appendix-C, contains sample calibration collection label Appendix-D contains sample data product label, and Appendix-E contain visualization of PDS4 data products.

1.4 Audience

This document serves both as a SIS and Interface Control Document (ICD). It describes both the archiving procedure and responsibilities, and data archive conventions and format. It is designed to be used both by the instrument data processing team in generating the archive, and by those wishing to understand the format and content of the NavCam PDS data product archive collection. Typically, these individuals would include scientists, data analysts, and software engineers.

1.5 Applicable Documents

- [1] Chandrayaan-3 SAC DP Design Review Document for Data Products Generation System and Archival (Chandrayaan-3/DP/SAC/SIPG/HRDPD/TR-05/Sep. 2022)

- [2] Chandrayaan-3 SAC DPGS and Archival ICD for MOX/ISSDC (Chandrayaan-3/DP/SAC/SIPG/HRDPD/TR-04/Apr 2023)

- [3] Handbook of Sensors (Navigational Camera) Chandrayaan-3, (LEOS-SSA-VSID-CHN3-DOC23-02-00 Issue 0 and Version 1)

- [4] Radiometric Characterization of Chandrayaan-3 Rover Navigational Camera (NavCam), (Chandrayaan-3/DP/SAC/SIPG/HRDPD/TR-06/Aug 2021)

- [5] Chandrayaan-3 Science Data Management and Archive Plan. Chandrayaan-3/HRDPD/SIPA/SAC/TR-16/September 2023

- [6] PDS4 Concepts Data Design Working Group October 1, 2018 version 1.11.0

- [7] Planetary Data System Standards Reference Version 1.11.0 October 1, 2018, Jet Propulsion Laboratory

- [8] The PDS4 Data Provider's Handbook Guide to Archiving Planetary Data Using the PDS4 Standard Version 1.11.0 October 1, 2018

- [9] PDS4 Data Dictionary Abridged – Version 1.11.0.0

- [10] List of Supplemental Formats (Approved by PDS Management Council: 2017-08-30)

- [11] Algorithm for Using CAHV to Determine SGI Graphics Viewpoint and Perspective, B. Bon, JPL IOM 3472-91-057, August 6, 1991.

1.6 Abbreviations Acronyms, and Description

Following **Table 1** brings out the list of definitions, acronyms & abbreviations used in this document.

Table 1 Acronym, Abbreviations and Description

Acronym	Abbreviations/Description
PDS	Planetary Data System
ISRO	Indian Space Research Organization
ISTRAC	ISRO Telemetry Tracking and Command Network
ISSDC	Indian Space Data Centre
ISDA	ISRO Science Data archive
SAC	Space Applications Centre
URSC	U.R. Rao Satellite Centre
LEOS	Laboratory for Electro Optics System
DP	Data Processing
NavCam	Navigational Camera
ICD	Interface Control Document
DPD	Deputy Project Director
CAHV	Center, Axis, Horizontal, Vertical (camera model)
Archive Team	Collection of teams from science (PI), Payload, Data Processing and ISSDC Operations

1.7 Glossary

Table 2 shows glossary of terms used in PDS4 domain.

Table 2 PDS4 Glossary

Terms	Definition
Archive	A place in which public records or historical documents are preserved; also the material preserved – often used in plural. The term may be capitalized when referring to all of PDS holdings – the PDS Archive.
Basic Product	The simplest product in PDS4; one or more data objects (and their description objects), which constitute (typically) a single observation, document, etc. The only PDS4 products that are not basic products are collection and bundle products
Bundle Product	A list of related collections. For example, a bundle could list a collection of raw data obtained by an instrument during its mission lifetime, a collection of the calibration products associated with the instrument, and a collection of all documentation relevant to the first two collections.
Class	The set of attributes (including a name and identifier) which describes an item defined in the PDS Information Model. A class is generic – a template from which individual items may be constructed.
Collection Product	A list of closely related basic products of a single type (e.g. observational data, browse, documents, etc.). A collection is itself a product (because it is simply a list, with its label), but it is not a basic product.
Data Object	A generic term for an object that is described by a description object. Data objects include both digital and non-digital objects.
Description Object	An object that describes another object. As appropriate, it will have structural and descriptive components. In PDS4 a ‘description object’ is a digital object – a string of bits with a predefined structure.
Digital Object	An object which consists of real electronically stored (digital) data.
Identifier	A unique character string by which a product, object, or other entity may be identified and located. Identifiers can be global, in which case they are unique across all of PDS (and its federation partners). A local identifier must be unique within a label.
Label	The aggregation of one or more description objects such that the aggregation describes a single PDS product. In the PDS4 implementation, labels are constructed using XML.<say what it contains>

Logical Identifier (LID)	An identifier which identifies the set of all versions of a product.
Versioned Logical Identifier (LIDVID)	The concatenation of a logical identifier with a version identifier, providing a unique identifier for each version of product.
Manifest	A list of contents.
Metadata	Data about data – for example, a ‘description object’ contains information (metadata) about an ‘object.’
Non-Digital Object	An object which does not consist of digital data. Non-digital objects include both physical objects like instruments, spacecraft, and planets, and non-physical objects like missions, and institutions. Non-digital objects are labeled in PDS in order to define a unique identifier (LID) by which they may be referenced across the system.
Object	A single instance of a class defined in the PDS Information Model.
PDS Information Model	The set of rules governing the structure and content of PDS metadata. While the Information Model (IM) has been implemented in XML for PDS4, the model itself is implementation independent.
Product	One or more tagged objects (digital, non-digital, or both) grouped together and having a single PDS-unique identifier. In the PDS4 implementation, the descriptions are combined into a single XML label. Although it may be possible to locate individual objects within PDS (and to find specific bit strings within digital objects), PDS4 defines ‘products’ to be the smallest granular unit of addressable data within its complete holdings.
Tagged Object	An entity categorized by the PDS Information Model, and described by a PDS label.
Registry	A data base that provides services for sharing content and metadata.
Repository	A place, room, or container where something is deposited or stored (often for safety).
XML	eXtensible Markup Language.
XML schema	The definition of an XML document, specifying required and optional XML elements, their order, and parent-child relationships.

2. Mission Overview

Chandrayaan-3 is the third spacecraft in its series of Lunar exploration. It carries Propulsion Module (PM), Lander Module (LM) and Rover. Chandrayaan-3 composite module mass of 3900 kg, got injected into the Earth Parking Orbit (EPO) of 170 x 36505 (nominal) dispersion ± 500 km with command cut-off. In EPO, several Earth-Bound Maneuvers (EBN) were performed to increase the apogee of the orbit and the combined stack is injected into the Lunar Transfer Trajectory (LTT).

The Propulsion Module carried the combined stack to moon through a Lunar Orbit Insertion (LOI) maneuver. It is shown pictorially in Figure1. Subsequently, the combined stack was lowered into a lunar orbit of 125 km x 160km with Lunar Burns (LBNs).

The Chandrayaan-3 mission carried PM & LM as composite until reaching Moon Bound Orbit of 125 km x 160km. Further, Propulsion Module and Lander module got separated as two independent satellites at pre-defined time and remained in same orbit for about one day. Further a de-boost maneuver was performed on lander to attain a orbit of 100 km x 30 km. Lander remained in this orbit for ~6 days to verify and validate the performance of all sub-systems followed by powered descent.

The total maneuver strategy was designed such that the orbit plane contains the desired landing site at powered descent start. At Perilune point, the powered descent got initiated through an Autonomous Landing Sequencer (ALS) and achieved a soft landing on the specified landing site located near lunar southern polar region. The mission profile is shown in Figure-1.

2.1 Mission Objective

- Expanding the technologies inherited from Chandrayaan-2 spacecraft and Develop and Demonstrate' newer technologies that will be useful for future planetary missions.
- To design, realize and deploy Lander capable of soft landing on a specified lunar site and deploy a Rover to carry out in-situ analysis of chemicals.

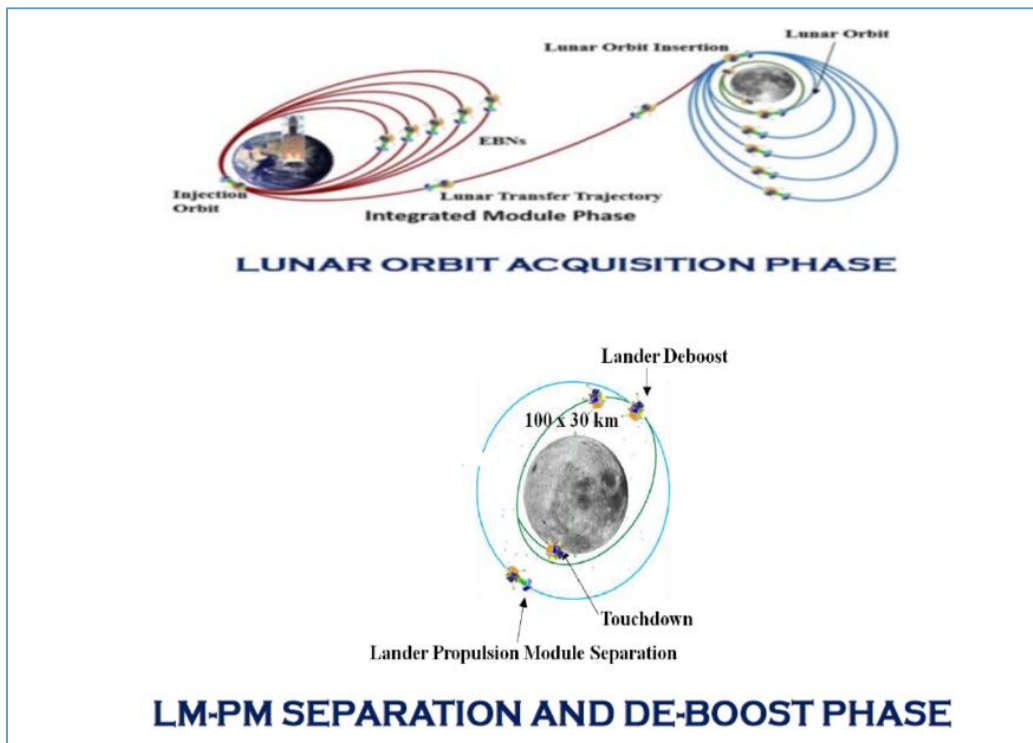
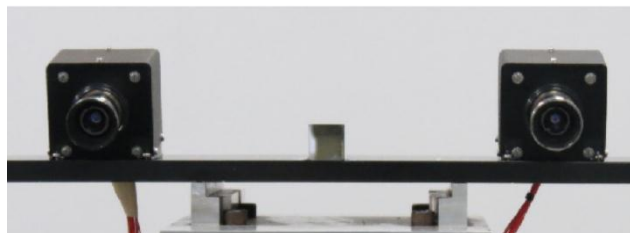


Figure 1 Mission Profile

2.2 Payload Objective

The Indian Rover landed on Moon through Chandrayaan-3 was a simple system to have the basic mobility and scientific operations on the lunar soil. The mobility of the Rover was planned to be semi-autonomous. A pair of Navigation Sensors was required to provide the imagery of the terrain ahead and a ground based image analysis and path planning system used these images to construct a three dimensional digital elevation model of the surface features imaged. The useful stereo based DEM generation and photogrammetric depth measurement demands that the field of views of both Navigation Sensors must overlap on each other for more than 60%. Hence the Navigation Sensors are to be oriented on the Rover deck in such a way that the above requirements are met. NAV Sensor has to detect 20mm object from 5-meter distance and power requirement should be less than 2 watts.



3. Instrument Overview

A pair of Rad Hard Stereo Vision Navigation Sensors was proposed for Vision based navigation and path planning for the Indian Lunar Rover on board Chandrayaan-3. These Navigation Sensors proposed to be mounted at the front face of the Rover, give information of obstacles like boulders/ craters/ pits on the path ahead. By analyzing the scene captured by these Navigation Sensors, the trajectory of the Rover was planned and executed. The figure shows the mounting location of Left and Right NavCam on the rover.

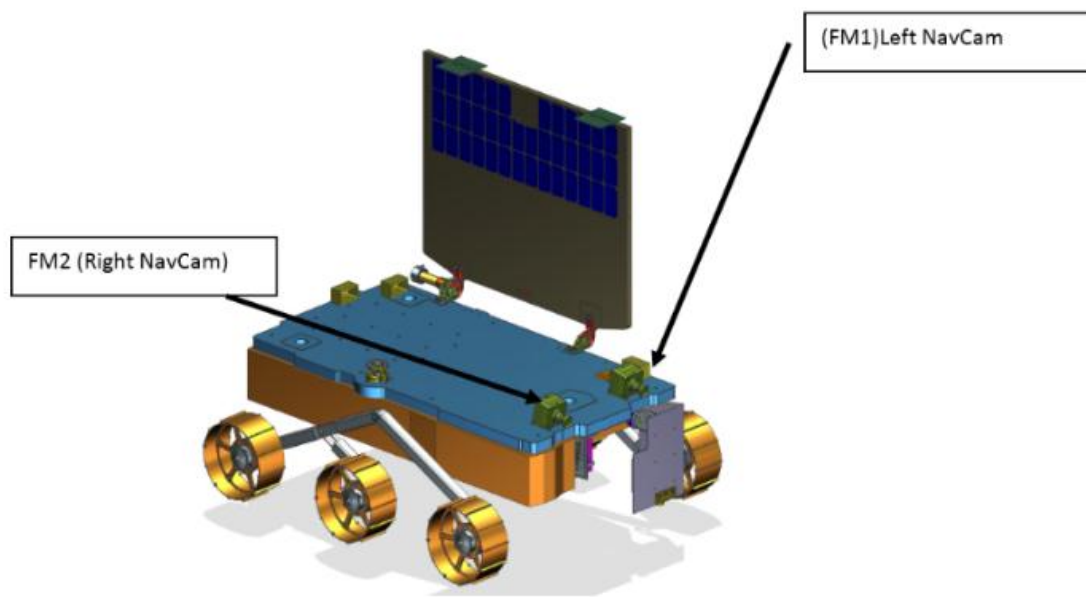


Figure 2 NavCam Left and Right on Rover

3.1 Specifications

The NavCam Payload specifications are tabulated in the table given below.

Table 3 Specification of NavCam

PARAMETER	VALUE
Detector	STAR 1000 APS (Monochrome)
No: of pixels	1024x1024
Pixel dimension	15 μm x 15 μm
Spectrum	Visible
Power	< 0.9 watt

Frame Rate	3 fps
Data rate	24 Mbps
Interface	Serial Digital (LVDS)
Image size on sensor	8.5 x 8.5 mm
Supply Voltage	5V $\pm$ 150mv
FOV	39° (H) x 39° (V)
Focal length	21 mm
Size of camera	55 x 50 x 75 mm(including lens projection)
Weight	180 gm including optics
Operating Temperature	-25 ° C to +65 ° C
Storage Temperature	-60 ° C to +70 ° C

3.2 Exposure Settings

The NavCam datasets were acquired by level-0 during the RPP operations at MOX. For every mobility data sets were acquired with gain 2 and three exposure settings. Three exposure settings were set in terms of integration lines:

- To avoid the getting dark image or oversaturated images.
- Due to terrain difference and presence of obstacle, local surface normal will be different which will make integration time setting as terrain specific.

Based on the sun illumination condition and NavCam viewing angle, the corresponding gain and exposure settings were selected and data was processed. The NavCams (left and right camera) were operated with gain as 2. During the initial days different exposure settings were attempted and later got finalized. The finalized exposure setting for the left image were the nominal integration line = 25 ($25 \times 0.35 = 8.75$ msec) and for the right image, the nominal integration line = 27 ($27 \times 0.35 = 9.45$ msec). Integration time is expressed as integration line which is configurable parameter got set during payload data acquisition. To derived the integration time in msec, integration line needs to be multiplied with 0.35 factor. Further more details about the gain and exposure settings done throughout the mission is tabulated under Appendix J (Rover NavCam Exposure Settings) and Appendix K (Rover NavCam Gain Settings).

4. Data Overview

This section provides a high level description of archive organization under the PDS4 Information Model (IM) as well as the flow of the data from the spacecraft through delivery to PDS. Unless specified elsewhere in this document, the Chandrayaan-3 NavCam archive confirms with version 1.11.0.0 of the PDS4 IM [4] and version 1.11 of the Chandrayaan-3 mission schema. A list of the XML Schema and Schematron documents associated with this archive are provided in **Table 4** below.

Table 4 Chandrayaan-3 NavCam Archive Schema and Schematron

XML Document	Steward	Product LID
PDS Master Schema, v.1.0.0.0	ISDA	urn:isro:isda:system_bundle:xml_schema:pds-xml_schema

4.1 Data Processing Levels

A number of different systems may be used to describe data processing level. This document refers to data by their PDS4 reduction level. Three different types of data products are envisaged for NavCam payload of Chandrayaan-3. **Table 5** provides a description of these levels along with the equivalent designations used in other systems.

Table 5 Chandrayaan-3 NavCam Data Processing Levels

PDS4 Level	PDS4 Level Description	ISRO Level	CODMAC Level
Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format.	0	2
Calibrated	Data converted to physical units, which makes values independent of the instrument.	1	3

4.2 Data Products

A PDS product consists of one or more digital and/or non-digital objects, and an accompanying PDS label file. Labeled digital objects are data products (i.e. electronically stored files). Labeled non-digital objects are physical and conceptual entities which have been described by a PDS label. PDS labels provide identification and description information for labeled objects. The PDS label defines a Logical Identifier (LID) by which any PDS labeled product is referenced throughout the system. In PDS4 labels are XML formatted ASCII files. More information on the formatting of PDS labels is provided in Section 6.3. More information on the usage of LIDs and the formation of Chandrayaan-3 LIDs is provided in Section 6.1. The PDS4 Data Products for NavCam defined based on the defined data processing levels (L0 and L1) as Raw and Calibrated data products. Table 6 shows the data products definitions for each levels of data processing. To visualize the data products please refer Annexure E. more details shown in next sub sections.

Table 6 Chandrayaan-3 NavCam Data Products Definitions

Data Products Levels	Level-0	Level-1
Data Products Definition	Raw data without any data processing correction	Radiometrically corrected image data
PDS4 Level	Raw	Calibrated

4.2.1 Raw Data Products

For each payload observation, PDS4 data products are generated (based on product id & sensor id derived from file name (section 6.5)) and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. Figure 4 shows structure of NavCam Raw Data Product file.

```

raw/20230828
├── ch3_nav_nrl_20230828T0555550000_d_img_d32_010.png
├── ch3_nav_nrl_20230828T0555550000_d_img_d32_010.xml
├── ch3_nav_nrl_20230828T1732040000_d_img_d32_013.png
├── ch3_nav_nrl_20230828T1732040000_d_img_d32_013.xml
├── ch3_nav_nrl_20230828T2026060000_d_img_gnh_014.png
├── ch3_nav_nrl_20230828T2026060000_d_img_gnh_014.xml
├── ch3_nav_nrr_20230828T0556380000_d_img_d32_010.png
├── ch3_nav_nrr_20230828T0556380000_d_img_d32_010.xml
├── ch3_nav_nrr_20230828T1732450000_d_img_d32_013.png
├── ch3_nav_nrr_20230828T1732450000_d_img_d32_013.xml
├── ch3_nav_nrr_20230828T2026470000_d_img_gnh_014.png
└── ch3_nav_nrr_20230828T2026470000_d_img_gnh_014.xml

```

Figure 3 NavCam Raw Data Archive

Raw Data Products contains raw data without any correction. **Table 7** shows the details of individual data products (with the data formats).

Table 7 NavCam Raw Data Products

PDS4 Data Product Levels	PDS4 Data Products	Data Formats	Description	Visualization tools available
Raw	Data Raw Image	PNG	The PDS Raw data in png format	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Data Raw Label	XML	The PDS Label Product contains description about the Raw Product.	PDS4 Viewer

4.2.2 Calibrated Data Products

For each payload observation, PDS4 data products are generated (based on product id derived from file name (section 6.5)), zipped and organized under year month day wise collection directory defined under instrument collection structure shown in section 6.4. Figure 4 shows unzipped structure of NavCam Calibrated Data Product zip file at user end.

```
calibrated/20230828
├── ch3_nav_ncl_20230828T0555550000_d_img_d32_010.png
├── ch3_nav_ncl_20230828T0555550000_d_img_d32_010.xml
├── ch3_nav_ncl_20230828T1732040000_d_img_d32_013.png
├── ch3_nav_ncl_20230828T1732040000_d_img_d32_013.xml
├── ch3_nav_ncl_20230828T2026060000_d_img_gnh_014.png
├── ch3_nav_ncl_20230828T2026060000_d_img_gnh_014.xml
├── ch3_nav_ncl_20230828T0556380000_d_img_d32_010.png
├── ch3_nav_ncl_20230828T0556380000_d_img_d32_010.xml
├── ch3_nav_ncl_20230828T1732450000_d_img_d32_013.png
├── ch3_nav_ncl_20230828T1732450000_d_img_d32_013.xml
├── ch3_nav_ncl_20230828T2026470000_d_img_gnh_014.png
├── ch3_nav_ncl_20230828T2026470000_d_img_gnh_014.xml
```

Figure 4 NavCam Calibrated Data Archive

Calibrated Data Products contains radiometrically corrected data along with the refined corner coordinates. **Table 8** shows the details of individual data products (with the data formats) present inside the zip product.

Table 8 NavCam Calibrated Data Products

PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Description	Visualization tools available
Calibrated	Data Calibrated Image	PNG	The PDS Radiometrically corrected data in png format	Any image viewer can be used to load the png file, so no need to load label file using PDS4 Viewer
	Data Calibrated Label	XML	The PDS Label Product contains description about the Calibrated Product.	PDS4 Viewer

4.3 Data Products Bundle

The NavCam data archive is organized into two bundles at data level. A description of each bundle is provided in **Table 9**. A more detailed description of the contents and format of each bundle is provided in Section 6.2.

Table 9 NavCam Data Products Bundles

Bundle Identifier	Logical	PDS4 Reduction Level	Description	Data Provider
urn:isro:isda:ch3_chr.nav.raw		Raw	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format	ISSDC
urn:isro:isda:ch3_chr.nav.calibrated		Calibrated	Data converted to physical units, which makes values independent of the instrument.	ISSDC

4.4 Data Flow

This section describes only those portions of the Chandrayaan-3 data flow from SAC POC to ISSDC that are directly connected to archiving. The ISSDC will maintain a long term archive of all Chandrayaan-3 science data, and be the sole long-term archive for all public Chandrayaan-3 data.

5. Archive Generation

The NavCam archive products are produced by the SAC DP team at SAC POC. The archive volume creation process will be described in this section. Archived data established at ISSDC by transferring the LTA data sets from the SAC POC, are made available to users electronically as soon as it qualifies as a science product but not later than lock in period after the delivery and validation of the data.

5.1 Data Processing and Production Software

After receiving Level 0 data from the level-0 software at SAC POC via NKN, SAC DP Software will take level-0 data as input and generate Level 0/1 raw and calibrated science data products. The science data products that the SAC DP Software delivers to the ISSDC will be stored by the ISSDC and will be made available to the PI team or as per the ISRO data distribution policy. The SAC POC will deliver archival-quality science data products to the ISDA for distribution to the public and long-term archiving in accordance with the NavCam Data Products & Archive SIS (this document).

5.2 Data Validation

There are two types of data validation. The first is validation of the science data and the second one is validation of the compliance of the archive with PDS archiving and distribution requirements. The first type of validation is carried out by the PI Team, and the second is overseen by the ISDA, in coordination with the PI Team. The formal validation of data content, adequacy of documentation, and adherence to PDS archiving and distribution requirements is scheduled and coordinated by the ISDA. This review includes peer review from the science community, at the ISDA's discretion. The review process may result in "liens," which are actions that are recommended by the reviewers or by PDS personnel to correct the archive. All liens must be resolved by the data-set provider. Once the liens are cleared, the ISDA completes a final validation before release.

5.3 Data Transfer Methods and Delivery Schedule

The Data Processing & Archival Software is running at SAC POC; products are generated at SAC POC. Initially active archive is established data will be collected as per mission operation plan. The products are available for the PI as initial validation of data. Table 10 shows NavCam data release schedule plan for Chandrayaan-3 mission.

Table 10 Data Release Schedule Plan

Payload	Release*	Data Collection Period (from date to date)	Lock-in-period (as per SSPO ISRO HQ)**	Date Released to Public***	Products
NavCam	First Version	24 Aug. 2023 to 02 Sept. 2023	6 month	April 2024	Raw and Calibrated

* Archive for the duration of Aug 2023 and Sept. 2023 will be released to the user.

** The lock in period duration will be required for initial data verification and validation and finally peer review process of the PDS data sets which will be going to release.

*** Release dates are dates that the ISDA releases Chandrayaan-3 archives to the public. Chandrayaan-3 delivers archives to the ISDA at least two weeks before the date of release to the public.

5.4 Data Product and Archive Volume Size Estimates

The total Archive Volume Size is worked out based on the data size, no. of orbits per day, mission life.

5.5 Backups and duplicates

The ISDA is responsible for maintaining at least three copies of its science archives and for delivering one copy of the data to the ISSDC for archive. As archives are released, ISSDC generate at least three copies on appropriate physical media for long-term storage.

6. Archive Organization and File Naming Conventions

6.1 Namespace Registration

PDS4 archive standard is adopted for Chandrayaan-3 mission. PDS4 is international archive standard being adopted by various space agencies across globe. There is International Planetary Data Alliance (IPDA), where all space agencies are members, ISRO is also one of them. IPDA has adopted PDS4. First step towards starting archive activities for any space agency is to register namespace via IPDA. ISRO has submitted namespace to IPDA. The namespace will be used as root to the archive of ISRO's upcoming and future planetary missions. For Chandrayaan-3 mission following name space will be used which is shown in the **Table 11**.

Table 11 ISDA Namespace for PDS4

Namespace ID	Logical Identifier prefix	Authority	Steward	Steward_ID	Provider/contact Point
isda	urn:isro:isda	ISRO Science Data Archive	ISRO	isda	Ajay Kumar Prashar Amitabh B.N. Ramakrishna

6.2 Logical Identifiers

Logical Identifier(LID) is unique to the product. Versioned Identifier (VID) indicates the versions improvements with respect to the products. All LIDs consist of a series of colon-separated segments.

Namespace:{BundleID}_{SubBundleID}:{CollectionID}_PDS4DataProductLevel:[{ProductID}]]. **Table 12** shows values for LID construction.

Table 12 LID for Chandrayaan-3 mission

Namespace	Bundle Id	Sub Bundle Id	Collection Id	PDS4 Data Product Level	Product Id
urn:isro:isda	ch3	chr	nav	raw calibrated	Address Data Products SIS document or EAICD

The following **table 13** shows the construction of LID to be used for all the scientific payloads of Chandrayaan-3 mission.

Table 13 LID for Chandrayaan-3 mission instruments

Instrument id	Logical Identifier
nav	urn:isro:isda:ch3_chr.nav:data_raw:productId urn:isro:isda:ch3_chr.nav:data_calibrated:productId

Products, collections, and bundles may evolve over time as calibration or other processing changes lead to improved versions. PDS appends a VID (version Identifiers) of the form M.n to LID to indicate different versions. The suffix is separated from LID by double colon. Archive structure is designed both at mission phase level and instrument level.

6.3 Mission Archive Structure at Bundle Level

At mission level, data is organized as mission id wise bundle. The mission id wise bundle at next level is organized based on the instrument host id wise bundle. The instrument host id bundle at next level is organized based on the mission phase id. Each mission phase id is having instrument id wise collection. All these mission attributes - mission id, instrument host id, mission phase id and instrument id are defined under archive conventions. For NavCam instrument, **table 14** shows values defined for mission attributes in archive conventions.

Table 14 Mission Attributes defined at structure level

Mission Attributes	Values	Defined at structure level
mission id	ch3	ch3
instrument host id	chr	chr
mission phase id	nop	nop (normal operation phase)
instrument id	nav	nav

All these attributes mentioned in above table are used in designing the archive structure at mission level. The figure 5 shows the mission level data products archive organization.

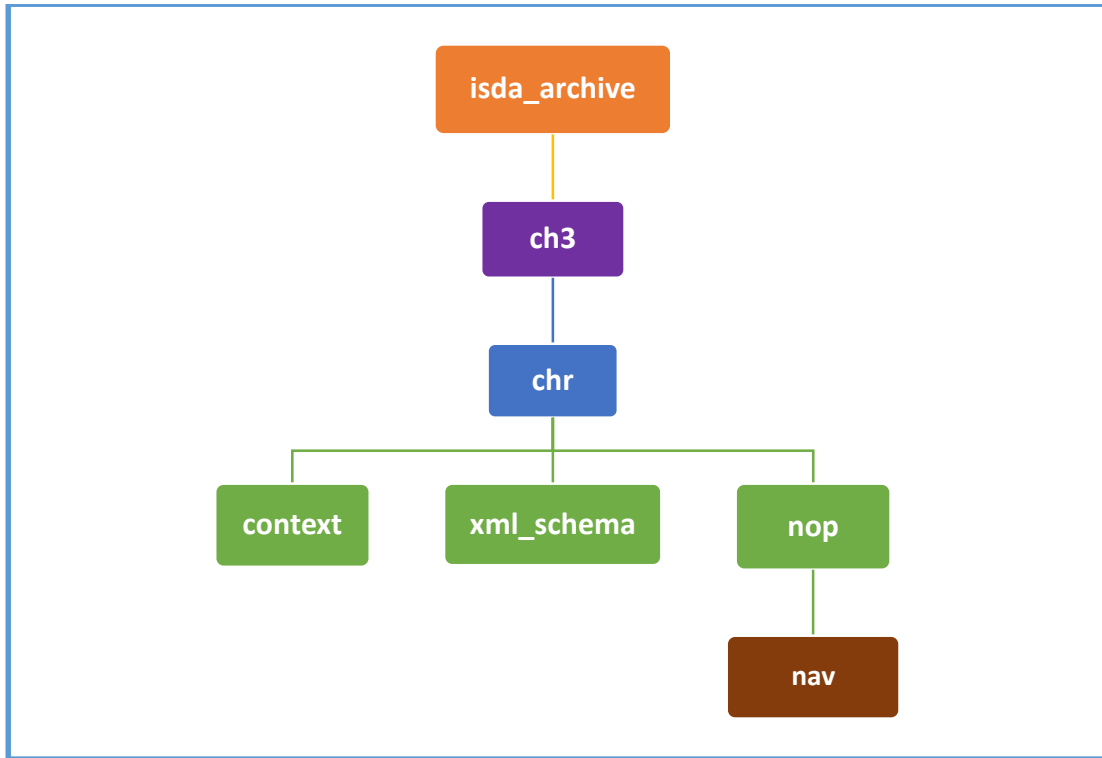


Figure 5 Chandrayaan-3 Mission Level Archive Organization

6.4 Instruments Collection

Under each instrument wise data collection, data is organized under each directory defined by PDS4. Data is organized based on the levels of data processing and data products definitions. Data products is organized into year, month and day wise as `yyyymmdd`. The year month and day wise directory contains PDS4 data products including PDS4 Label products. Figures 6 shows the detail of nav instrument wise collection structure.

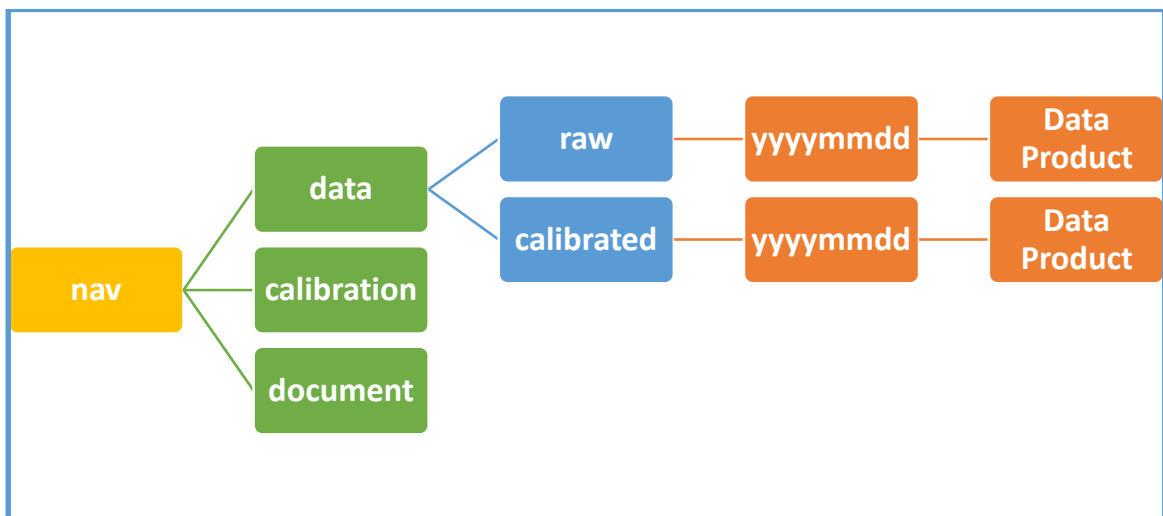


Figure 6 Chandrayaan-3 nav Instrument wise collection structure

At instrument level, collection of different product types is defined as per standard PDS4 conventions. Table 15 defined in detail about collection types.

Table 15 Collection product types

File or Directory Name	Description
root	Root directory describing full nav collection data sets
calibration	Contains data and files necessary for the calibration of basic products.
context	Contains list of products comprising various objects, identified with in the PDS4 registry, that are specific to the science bundle. These include physical objects such as instruments, spacecraft, and planets and conceptual objects such as missions.
document	Contains all components (figure, tables, text, etc.) of one or more related documents. The document include software interface specifications (SISs), calibration reports.
data	Contains observational products, often separated according to processing level, target, instrument mode, etc.
xml_schema	Contains XML schemas and related products which may be used for generating and validating PDS4 labels.

6.4.1 Root directory content

Files in the root directory include an overview of the archive, a description of the collection for the PDS4 data set. The following files are contained in the Root Directory shown in Table 16.

Table 16 Root directory content products

File Name	File Contents	File Provided By
bundle.xml	Collection content and format information	Archive Team
readme.txt	Collection content and format information	Archive Team

6.4.2 Calibration directory content

The calibration directory contains calibration files used to process the data products, or calibration data needed to use the data products. This calibration files are the look up table files that got generated during ground calibration exercise. All these lookup table files naming convention is based on the parameters - mission (ch3), instrument_id (nav), image (left/right), gain (gain), offset (offset) and gain offset (gain_offset). Apart from Look up table files, two more files got added as part of CAHV

camera model based on the file naming convention which is based on the parameters - mission (ch3), instrument_id (nav), image (left/right), CAHV (cahv). For more details on CAHV please refer Appendix J. The following files are contained in the calibration directory shown in **table 17**.

Table 17 Calibration directory content products

File Name	File Contents	File Provided By
collection_calibration_inventory.csv	Contains about the list of calibration data products in the collection inventory	Archive Team
collection_calibration_inventory.xml	XML Label defining the inventory csv file	Archive Team
Calibration files (Look Up Table)	Look up table files (ch3_nav_left_offset.txt, ch3_nav_right_offset.txt, ch3_nav_left_gain.txt, ch3_nav_right_gain.txt, ch3_nav_left_gain_offset.txt, ch3_nav_right_gain_offset.txt)	Archive Team
Calibration xml files (Look Up Table)	XML label defining the look up table	Archive Team
Calibration files (camera model - CAHV)	Camera Model CAHV files (ch3_nav_left_cahv.txt, ch3_nav_right_cahv.txt)	Archive Team
Calibration xml files (camera model - CAHV)	XML label defining the cahv files	Archive Team

6.4.3 Context directory content

The context directory contains the list of products comprising various objects, identified with in the PDS4 registry, that are specific to the science bundle. These include physical objects such as instruments, spacecraft, and planets and conceptual objects such as missions. The following files are contained in the context directory shown in **table 18**.

Table 18 Context directory content products

File Name	File Contents	File Provided By
collection_context_inventory.csv	Contains about the list of context data products in the collection inventory	Archive Team
collection_context_inventory.xml	XML Label defining the inventory csv file	Archive Team

6.4.4 Document directory content

The document directory contains documentation to help the user understand and use the archive data. The following files are contained in the document directory shown in the **table 19**.

Table 19 Document directory content products

File Name	File Contents	File Provided By
collection_document_inventory.xml	Label to define the inventory	Archive Team
collection_document_inventory.csv	A description of the contents of this directory	Archive Team
ch3_nav_pds_dp_archive_sis.pdf	Description of instrument and PDS data products and Archive Structure	Archive Team
ch3_nav_pds_dp_archive_sis.xml	XML label defining nav pds sis pdf	Archive Team

6.4.5 Data directory content

Under the data directory there are three separate subdirectory based on the defined PDS4 data products levels – raw and calibrated. Under each sub directories, there are again sub directory based on year, month and day wise. The year, month and day wise subdirectory contains PDS4 data products mainly image and associated label files in XML format. All the browse binary images are named as per file naming conventions mentioned under section 6.5.

6.4.6 Xml_schema directory content

This directory contains XML schemas and related products which may be used for generating and validating PDS4 labels. The following files are contained in the xml_schema directory shown in **table 20**.

Table 20 Xml_schema directory content products

File Name	File Contents	File Provided By
collection _xml_schema_inventory.csv	Contains about the list of xml schema in the collection inventory	Archive Team
collection _xml_schema_inventory.xml	XML Label defining the xml schema csv file	Archive Team

6.5 File Naming Conventions & Formats

This section describes the naming convention used for science data files for the Chandrayaan-3 mission. **Table 22** shows the detail description about the following file naming conventions designed for the data products.

ch3_<nav>_<mtc>_<YYYYMMDDTHHMMSSsss>_<p>_<prd>_<stn>_mob.fff

Table 21 File Naming Conventions

Code	Description
ch3	It denotes the Mission name whose value is fixed as ch3
nav	It denotes the Instrument id whose value is fixed as nav
m	It denotes the mission phase name, it can have following value = e (refers to the Earth Bound Phase - ebp) = n (refers to the Normal Operations Phase - nop)
t	It denotes the data type; it can have following values: = r (it denotes raw data) = c (It denotes calibrated data) = d (It denotes derived data)
c	It denotes the imaging mode/camera id/band. It can be one of the following: = l (it denotes left camera of NavCam) = r (it denotes right camera of NavCam)
YYYYMMDDTHHMMSSsss	It denotes the observation/imaging start time in format [YYYYMMDDTHHMMSSsss] YYYY = stand for year MM = stand for month in MM format DD = stand for day in DD format T = stand for time. HH = stand for hour in HH format MM = stand for minute in MM format SSS = stand for seconds in SS format ssss = stand for nanosecond in ssss format
p	It denotes the PDS Data Products categories. It value can be any of the following: = d (It denotes data products under data directory) = b (it denotes browse data products under browse directory) = g (it denotes gridded data products under geometry directory)

prd	It denotes the PDS Data Product Name. It value can be any of the following: = img (It denotes the image data product) = brw (It denotes the browse data product) = grd (It denotes the gridded data product) = dtm (it denotes DTM derived product) = orh (it denotes Ortho derived product)
stn	It denotes the station id from which data was acquired. It value can be any of the following: = d32 (ISSDC Bangalore) = d18 (ISSDC Bangalore) = gds (Gold Stone USA) = cnb (Canberra, Australia)
mob	Rover mobility id given by the mission team
fff	It denotes the standard file type extension. It can have any one of the following values: = png (PDS Image Data File) = xml (PDS Detached Label File) = png (PDS Browse Data File) = csv (PDS Geometry Data File) = tif (PDS Geo tiff data)

Below table shows sample file naming, data formats for the instruments mentioned in the above table.

Table 22 PDS4 File Names and Data Formats

Instrument	File Name	Data Formats
NAV Left	ch3_nav_ncl_20190628T1645300000_d_img_d32_001.png	PNG
	ch3_nav_ncl_20190628T1645300000_d_img_d32_001.xml	XML
NAV Right	ch3_nav_ncr_20190628T1645300000_d_img_d32_001.png	PNG
	ch3_nav_ncr_20190628T1645300000_d_img_d32_001.xml	XML

7. Archive Products formats

Data that comprises the NavCam archives are formatted in accordance with PDS specifications. This section provides details on the formats used for each of the products included in the archive. The detailed file formats for the defined data products based on PDS4 data types have been defined shown in next sub section.

7.1 Data File Formats

Nav instrument is an imaging payload. The primary data collected from instrument is in image format. The primary data collected from instrument is raw image. It is a panchromatic camera. Both primary as well as secondary data sets are defined that contains data product levels, types and formats shown in the **table 23**.

Table 23 NavCam PDS4 Data Products

Sensors	PDS4 Data Product Levels	PDS4 Data Products Types	Data Formats	Data type*
Panchromatic	Raw	Image	PNG	UnsignedByte
	Calibrated	Image	PNG	UnsignedByte

Note*:

Unsigned Byte is 8 bit

7.2 Document Product File Formats

All the files under document collection will be written in text format, csv, png, xml and pdf

7.3 PDS Labels

PDS labels are ASCII text files written, in the eXtensible Markup Language (XML). All product labels are detached from the digital files (if any) containing the data objects they describe (except Product_Bundle). There is one label for every product. Each product, however, may contain one more data objects. The data objects of a given product may all reside in a single file, or they may be stored in multiple separate files. PDS4 label files must end with the file extension “.xml”.

7.3.1 XML Documents

For the Chandrayaan-3 mission PDS labels will conform to the PDS master based schema based upon the 1.11.0.0 version of the PDS Information Model for structure, and the 1.11.0.0 version of the PDS schematron for content. By use of an XML editor

these documents may be used to validate the structure and content of the product labels.

The PDS master schema and schematron documents are produced, managed, and supplied to Chandrayaan-3 by the ISDA Archive Working Group. In addition to these documents, the Chandrayaan-3 Archive Working Group has produced Chandrayaan-3 Archive conventions containing the additional XML attribute and parameter definitions specific to Chandrayaan-3 mission.

Examples of PDS labels required for the NavCam archive are shown in Appendix B (bundle products), Appendix C (collection products), Appendix D (basic products), Appendix E (Local Data Dictionary).

7.4 Delivery Package

Data transfers, from ISDA to the data users, are accomplished using delivery packages. Delivery packages include the following required elements:

- The package which consists of a compressed bundle of the products being transferred.
- A transfer manifest which maps each product's LID/VID to the physical location of the product label in the package after uncompressing.
- A checksum manifest which lists the MD5 checksum of each file included in the package after uncompressing.

Appendix A Support staff and cognizant Persons

Table 24 Chandrayaan-3 Team staff

Name	Contact Person	Contact	Email Id
Indian Space Science Data Center IDSN Bylalu, Bengaluru	ISSDC team	+91-80-2202-9566/9567	issdc@istrac.gov.in
Space Applications Centre (SAC), Ahmedabad	Data Processing (DP) Team	+91-79-26914727/63/64	amitabh@sac.isro.gov.in

Appendix B Sample Bundle Product Label

```
<Product_Bundle xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
https://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1700.xsd">
  <Identification_Area>
    <logical_identifier>urn:isro:isda:ch3_chr</logical_identifier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-3 Rover NavCam Data Archive Products Bundle</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Bundle</product_class>
    <Citation_Information>
      <author_list>isro,issdc,sac</author_list>
      <publication_year>2024</publication_year>
      <description>Chandrayaan-3 Rover NavCam Payload Data Products</description>
    </Citation_Information>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2023-11-02</modification_date>
        <version_id>1.0</version_id>
        <description>Peer reviewed version of the Chandrayaan-3 Rover NavCam payload data
products bundle</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Context_Area>
    <Time_Coordinates>
      <start_date_time>2023-08-24Z</start_date_time>
      <stop_date_time>2023-09-02Z</stop_date_time>
    </Time_Coordinates>
    <Investigation_Area>
      <name>Chandrayaan-3</name>
      <type>Mission</type>
      <Internal_Reference>
        <lid_reference>urn:isro:isda:ch3_chr.nav:context:mission</lid_reference>
        <reference_type>bundle_to_investigation</reference_type>
      </Internal_Reference>
    </Investigation_Area>
    <Observing_System>
      <name>Chandrayaan-3</name>
      <Observing_System_Component>
        <name>Chandrayaan-3 Rover</name>
        <type>Spacecraft</type>
        <Internal_Reference>
          <lid_reference>urn:isro:isda:ch3_chr.nav:context:inst_host_spacecraft</lid_reference>
          <reference_type>is_instrument_host</reference_type>
        </Internal_Reference>
      </Observing_System_Component>
      <Observing_System_Component>
        <name>orbiter high resolution camera</name>
        <type>Instrument</type>
```

```

    <Internal_Reference>
      <lid_reference>urn:isro:isda:ch3_chr.nav:context:inst_nav</lid_reference>
      <reference_type>is_instrument</reference_type>
    </Internal_Reference>
  </Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <Internal_Reference>
    <lid_reference>urn:isro:isda:context:target:satellite.moon</lid_reference>
    <reference_type>data_to_target</reference_type>
  </Internal_Reference>
</Target_Identification>
</Context_Area>
<Bundle>
  <bundle_type>Archive</bundle_type>
</Bundle>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch3_chr.nav:calibration</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_calibration_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch3_chr.nav:data</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_data_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch3_chr.nav:document</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_document_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch3_chr.nav:context</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_context_collection</reference_type>
</Bundle_Member_Entry>
<Bundle_Member_Entry>
  <lid_reference>urn:isro:isda:ch3_chr.nav:schema</lid_reference>
  <member_status>Primary</member_status>
  <reference_type>bundle_has_schema_collection</reference_type>
</Bundle_Member_Entry>
</Product_Bundle>

```

Appendix C Sample Collection Product Label

```
<Product_Collection xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1400.xsd
http://pds.nasa.gov/pds4/disp/v1
http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1
http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd">

<Identification_Area>
  <logical_identifier>urn:isro:isda:ch3_chr.nav:document</logical_identifier>
  <version_id>1.0</version_id>
  <title>Chandrayaan-3 Rover NavCam document file collection</title>
  <information_model_version>1.11.0.0</information_model_version>
  <product_class>Product_Collection</product_class>
  <Citation_Information>
    <author_list>Chandrayaan-3 DP Team</author_list>
    <editor_list>Chandrayaan-3 DP Team</editor_list>
    <publication_year>2024</publication_year>
    <description>collection of document products.</description>
  </Citation_Information>
  <Modification_History>
    <Modification_Detail>
      <modification_date>2023-11-02</modification_date>
      <version_id>1.0</version_id>
      <description>Initial version</description>
    </Modification_Detail>
  </Modification_History>
</Identification_Area>
<Context_Area>
  <Investigation_Area>
    <name>Chandrayaan-3</name>
    <type>Mission</type>
    <Internal_Reference>

<lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan3::1.0</lid
vid_reference>
    <reference_type>collection_to_investigation</reference_type>
  </Internal_Reference>
</Investigation_Area>
  <Observing_System>
    <Observing_System_Component>
      <name> Chandrayaan-3 Rover Module </name>
      <type>Spacecraft</type>
      <description>
        Chandrayaan-3 Rover Module is a rover craft consists of various scientific
```

instruments.

```
</description>
</Observing_System_Component>
<Observing_System_Component>
  <name>orbiter high resolution camera</name>
  <type>Instrument</type>
  <description>
    Navigational Camera (NavCam) is one of the engineering instrument hosted
    on the Chandrayaan-3 Rover Module. This instrument is used for rover mobility
  </description>
</Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <description>Moon is a natural satellite of Earth</description>
</Target_Identification>
</Context_Area>
<Collection>
  <collection_type>Document</collection_type>
</Collection>
<File_Area_Inventory>
  <File>
    <file_name>collection_document_inventory.csv</file_name>
    <local_identifier>collection_document_inventory.csv</local_identifier>
    <creation_date_time>2020-07-28</creation_date_time>
    <file_size unit="byte">73</file_size>
    <records>1</records>
    <md5_checksum>09a718293d7ee74abaafa2722619865f</md5_checksum>
  </File>
  <Inventory>
    <offset unit="byte">0</offset>
    <parsing_standard_id>PDS DSV 1</parsing_standard_id>
    <records>1</records>
    <record_delimiter>Carriage-Return Line-Feed</record_delimiter>
    <field_delimiter>Comma</field_delimiter>
    <Record_Delimited>
      <fields>2</fields>
      <groups>0</groups>
      <maximum_record_length unit="byte">259</maximum_record_length>
      <Field_Delimited>
        <name>Member Status</name>
        <field_number>1</field_number>
        <data_type>ASCII_String</data_type>
        <maximum_field_length unit="byte">1</maximum_field_length>
        <description>This column specifies the member status</description>
      </Field_Delimited>
```

```
<Field_Delimited>
  <name>LIDVID_LID</name>
  <field_number>2</field_number>
  <data_type>ASCII_LIDVID_LID</data_type>
  <maximum_field_length unit="byte">255</maximum_field_length>
  <description>This column specifies the LIDVID of the files that comprise the
collection.</description>
</Field_Delimited>
</Record_Delimited>
<reference_type>inventory_has_member_product</reference_type>
</Inventory>
</File_Area_Inventory>

</Product_Collection>
```

Appendix D Sample Data Product Labels

```
<Product_Ancillary xmlns="http://pds.nasa.gov/pds4/pds/v1"
xmlns:isda="http://pds.nasa.gov/pds4/isda/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1B00.xsd
http://pds.nasa.gov/pds4/disp/v1
http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1 http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd
https://isda.isdc.gov.in/pds4/isda/v1
https://isda.isdc.gov.in/pds4/isda/v1/ch3_ldd_ISDA_1000.xsd">

  <Identification_Area>

    <logical_identifier>urn:isro:isda:ch3_chr.nav:data_raw:ch3_nav_nrl_20230825t1040
153807_d_img_d32_003</logical_identifier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-3 Rover NavCam Mobility Experiment</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Ancillary</product_class>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2024-06-11</modification_date>
        <version_id>1.0</version_id>
        <description>PDS4 product label created by SAC DP Team</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Context_Area>
    <Time_Coordinates>
      <start_date_time>2023-08-25T10:40:15.3807Z</start_date_time>
      <stop_date_time>2023-08-25T10:40:16.1058Z</stop_date_time>
    </Time_Coordinates>
    <Primary_Result_Summary>
      <purpose>Navigation</purpose>
      <processing_level>Raw</processing_level>
      <description>Raw Processing Level describes the raw data without any correction
applied.</description>
    </Primary_Result_Summary>
    <Investigation_Area>
      <name>Chandrayaan-3</name>
      <type>Mission</type>
      <Internal_Reference>

      <lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan3::1.0</lid
vid_reference>
```

```

    <reference_type>data_to_investigation</reference_type>
  </Internal_Reference>
</Investigation_Area>
<Observing_System>
  <Observing_System_Component>
    <name>Chandrayaan-3 Rover Module</name>
    <type>Spacecraft</type>
    <description>
      Chandrayaan-3 Rover module is a rover craft consists of various scientific
instruments.
    </description>
  </Observing_System_Component>
  <Observing_System_Component>
    <name>Navigational Camera</name>
    <type>Instrument</type>
    <description>
      Navigational Camera (NavCam) is one of the engineering instrument hosted
      on the Chandrayaan-3 Rover Module. This instrument is used for rover
mobility
      on lunar surface.
    </description>
  </Observing_System_Component>
</Observing_System>
<Target_Identification>
  <name>Moon</name>
  <type>Satellite</type>
  <description>Moon is a natural satellite of Earth</description>
</Target_Identification>
<Mission_Area>
  <isda:Product_Parameters>

<isda:job_id>NCBXXD32C3R0018000NNNN23237113106000_V2_1</isda:job_id>
  <isda:dumping_orbit_number>180</isda:dumping_orbit_number>
  <isda:dop>20230825</isda:dop>
  <isda:frame_number>3</isda:frame_number>
  <isda:current_mobility_id>002</isda:current_mobility_id>
  <isda:future_mobility_id>003</isda:future_mobility_id>
  <isda:operation_id>000001</isda:operation_id>
  <isda:day_no>237</isda:day_no>
  <isda:year>2023</isda:year>

<isda:inclinometer_utc>2023:237:10:49:00:7082023:237:10:49:00:578</isda:inclino
meter_utc>
  <isda:inclinometer_x_angle>-0.907</isda:inclinometer_x_angle>
  <isda:inclinometer_y_angle>-4.672</isda:inclinometer_y_angle>
  <isda:inclinometer_roll_angle>0.0</isda:inclinometer_roll_angle>
  <isda:inclinometer_pitch_angle>0.0</isda:inclinometer_pitch_angle>

```

```

    <isda:sensor_id>left camera</isda:sensor_id>
    <isda:gain>2</isda:gain>
    <isda:integration_time>20</isda:integration_time>
  </isda:Product_Parameters>
</Mission_Area>
</Context_Area>
<File_Area_Ancillary>
  <File>

    <file_name>ch3_nav_nrl_20230825T1040153807_d_img_d32_003.png</file_name>
    <creation_date_time>2024-06-12T15:36:55</creation_date_time>
    <file_size unit="byte">538421</file_size>
    <md5_checksum>981e24e50057fc4c1b5cc06196fe5550</md5_checksum>
    <comment>
      This File contains the count raw image in png format.
      md5_checksum is provided for ensuring data integrity when users are
      downloading the data.
    </comment>
  </File>
  <Encoded_Image>
    <name>ch3_nav_nrl_20230825T1040153807_d_img_d32_003</name>
    <offset unit="byte">0</offset>
    <encoding_standard_id>PNG</encoding_standard_id>
  </Encoded_Image>
</File_Area_Ancillary>
</Product_Ancillary>

```

Appendix E Local Data Dictionary

Every mission has some conditions based on the events that needs to be handle as an information at output product level. Some information is common to every planetary mission launched by ISRO. There is an ISRO Science Data Archive established during Chandrayaan-1 mission in 2008. All the products hosted for users under ISDA are PDS compliance. For Chandrayaan-3 mission, PDS4 Archive standard was adopted. PDS4 is currently next and future generation archive standard adopted by all the space agencies including NASA, ESA and JAXA for their current and future mission. To handle the ISRO planetary mission's specific parameters, there is a concept in PDS4 called Local Data Dictionary i.e. LDD. In LDD one can define all the parameters and that can be used as a reference for future planetary missions in India. LDD takes basic data dictionary of PDS4 and then top of it builds local dictionary. For Chandrayaan-3 mission and future planetary missions of ISRO, list of parameters is identified shown in details in table 25.

Table 25 ISDA Local Data Dictionary Class and Attributes List

Class/Sub Class Hierarchy	Attribute Name	Description/Value
Mission_Area Product_Parameters	job_id	Unique id assign by DP Scheduler when the product is ingested
	dumping_orbit_number	Dumping Orbit Number is assigned based on the data collected on ground station
	dop	Date of Pass on which data got acquired
	frame_number	Frame number indicates selected frame out of three frames acquired during data acquisition
	current_mobility_id	Current rover mobility id assigned during the RPP Operations
	future_mobility_id	Future rover mobility id assigned during the RPP Operations
	operation_id	Unique identifier assigned during RPP Operations
	day_no	Julian day number on which RPP Operations carried out
	year	year on which RPP Operations carried out
	inclinometer_utc	Inclinometer time reading as per UTC format
	inclinometer_x_angle	Inclinometer x angle

	inclinometer_y_angle	Inclinometer y angle
	inclinometer_roll_angle	Inclinometer Roll angle if not available value will be n/a means not available
	inclinometer_pitch_angle	Inclinometer Pitch angle if not available value will be n/a means not available
	gain	Angle between the projection of sun rays and a line due south or north and value is expressed in degree
	integration_time	Integration time is expressed as integration line which is configurable parameter got set during payload data acquisition. To derived the integration time in msec, integration line needs to be multiplied with 0.35 factor

Appendix F Data Disclaimer

The Rover NavCam was the engineering payload consisting of left camera and right camera. The main payload objective was to enable the rover movement on the moon surface. The payload was operated based on different gain and exposure settings and acquired the left camera image and right image. For some of the data products (left and right image) as expected saturation was observed due to the sun illumination condition and NavCam viewing angle.

It is to bring to the notice of the user that the Department of Space/Indian Space Research Organization (DOS/ISRO) is not responsible for any dispute/disrepute that may arise due to the use of this dataset. The user should neither redistribute the data to third party nor it be used for any commercial purpose without the knowledge of Department of Space/Indian Space Research Organization (DOS/ISRO).

Appendix-G: Raw to Calibrated Data Conversion

The step wise procedure for converting raw image data to calibrated image data image raw count to rad cunt is as follows:

- (1) **Raw Count Data Extraction** - Read the raw data (left camera or right camera) get the raw count data. Subtract the offset from the raw count data using following formula.

Raw count = Raw Image Count - offset

Where,

Raw Image Count is raw image pixel value

Offset is Look up table under calibration directory (table 17) based on the left or right camera data processing.

- ch3_nav_left_offset.txt
- ch3_nav_right_offset.txt

- (2) **Get Raw Minimum and Maximum Pixel Value** - After the subtraction (step 1), get the minimum and maximum count value stored as **raw_min** and **raw_max**.

- (3) **LUT Application** - Look up table is applied to the Raw Count data using following formula

Radiance value = Raw count * multiplicative factor (gain) + gain_offset

where,

Raw Count is the raw image data values obtained from step (1),

multiplicative factor (gain) is obtained from the Look up table based on left or right camera data processing,

- ch3_nav_left_gain.txt
- ch3_nav_right_gain.txt

gain_offset is obtained from the Look up table based on the left or right camera data processing,

- ch3_nav_left_gain_offset.txt
- ch3_nav_right_gain_offset.txt

- (4) **Get Rad Minimum and Maximum Pixel Value** - after the LUT application (step 3), get the minimum and maximum count value stored as **rad_min** and **rad_max**.

- (5) **Radiance Data Generation** – Final step the radiance data is generated using following formula

Radiance value = Radiance value * (raw_max - raw_min) / (rad_max - rad_min) where,

Radiance value is the rad image data values obtained from step (3),

raw_max is from step (2), **raw_min** is from step (2)

rad_max is from step (4), **rad_min** is from step (4)

Appendix-H: CAHV Camera Model

The CAHV model is a linear, perspective-projection camera model (equivalent to a pinhole camera). It consists of four 3-vectors (C,A,H,V) that describe the internal and external camera model parameters needed to translate between 2D image coordinates and 3D world coordinates. C (Vector_Center) is the 3D position of the pinhole (center of the entrance pupil). A (Vector_Axis) is a unit vector normal to the image plane pointing outward. H (Vector_Horizontal) is a composite vector encoding three quantities: H' (a vector in the image plane perpendicular to the vertical columns), Hs (the distance between the lens center and image plane, measured in horizontal pixels), and Hc (the horizontal image coordinate directly under C when moving parallel to A). V (Vector_Vertical) similarly composites the analogous V', Vs, and Vc in the vertical direction.

C - The 3D position of the entrance pupil.

A - A unit vector normal to the image plane pointing outward (towards C).

H - A vector pointing roughly rightward in the image; it is a composite of the orientation of the CCD rows, the horizontal scale, the horizontal center.

V - A vector pointing roughly downward in the image; it is a composite of the orientation of the CCD columns, the vertical scale, the vertical center, and A.

If P is a point in the scene, then the corresponding image locations x and y can be computed from:

$$x = \frac{(P-C) \cdot H}{(P-C) \cdot A}$$

$$y = \frac{(P-C) \cdot V}{(P-C) \cdot A}$$

For details on the camera model math and calibration and more description of the CAHV-model family, see references [Ref 11].

Appendix-I: Rover Mobility Summary

Timeline (Mobility)	MID	Path	Description
23.08.2023 236:19:06:36	001	2.3m on 12deg ramp, 0.7m	Roll down
24.08.2023 236:21:03:52	002	2.499m, 1deg CW, 1.587m, 45.138deg CCW	
25.08.2023 237:18:59:06	003	2.76m, 12.713deg CW, 2.283m, 24.465deg CCW	
26.08.2023 238:17:43:00	004	30deg CCW	VO=31.5 deg, Slip=(40-31.5)/40=21.25%
27.08.2023 239:03:40:00	005	25deg CCW, 25deg CW	Decided to retrace because of crater
27.08.2023 239:16:00:50	006	30deg CW	
27.08.2023 239:16:20:54	007	24.465deg CW, 2.283m Rev., 12.713deg CCW, 2.76m Rev.	Retrace of MID3
27.08.2023 239:21:04:11	008	36deg CCW	
28.08.2023 240:05:39:13	009	2..754m, 12.168deg CCW, 1.258m, 12.168deg CW	4 mm DEM
	010		Commanding issue
	011		Commanding issue
28.08.2023 240:17:15:00	012	2.754 m Rev., 12.168deg CCW, 1.758m, 12.168 ⁰ CW	Retrace of MID9 with overshoot of 0.5m towards west to avoid Lander shadow
28.08.2023 240:20:20:09	013	0.5m Rev.	
29.08.2023 241:05:20:00	014	180deg CW, 0.5m, 90deg CW,90deg CW	
29.08.2023 241:08:39:30	015	35 deg CW	To compensate for slip during MID14
29.08.2023 241:15:43:00	016	1m Rev.	Movement w/o DEM based on LI image
29.08.2023 241:16:37:00	017	1m Rev.	Movement w/o DEM based on LI image
29.08.2023 241:16:45:00	018	1m Rev.	Movement w/o DEM based on LI image
29.08.2023 241:16:58:30	019	1m Rev.	Movement w/o DEM based on LI image
29.08.2023 241:18:46:03	020	180 deg CW, 0.5 m, 180 deg CCW	Imaging for movement towards west crater
29.08.2023 241:21:51:00	021	2.587m Rev. 16.361deg CW, 3.557m Rev. 16.361deg CCW	Reverse movement based on MID20 image
29.08.2023 241:23:18:00	022	180 deg CW (NavCam),0.5m, 180deg CCW	Imaging for movement towards west crater
30.08.2023 242:01:58:00	023	30deg CCW (NavCam), 30deg CW	To capture Lander
30.08.2023	024	2.933m Rev. 2.231 deg	Reverse mobility based on MID22

242:03:14:00		CW, 2.568m Rev., 2.231 deg CCW	image
30.08.2023 242:04:06:00	025	180deg CW, 0.5m, 180deg CCW (NavCam)	Imaging for movement towards west crater
30.08.2023 242:04:59:00	026	20deg CCW, 20deg CW	For Lander Pic-2
30.08.2023 242:10:43:00	027	2.456m Rev, 8.031deg CCW, 3.579m Rev., 8.031deg CCW	Reverse movement based on MID25 image
30.08.2023 242:11:38:30	028	170deg CW, 0.5m, 170deg CCW	Imaging for movement towards west crater
30.08.2023 242:14:09:00	029	2.591m Rev., 16.604deg CW, 2.379m Rev., 39.812deg CCW, 2.382m Rev., 10.975 deg CCW	Reverse movement based on MID28 image
30.08.2023 242:14:37:00	030	180deg CW, 0.5m Rev., 180deg CCW	Imaging for movement towards west crater
30.08.2023 242:18:36:00	031	2.545m Rev., 8.381deg CW, 2.037m Rev., 23.724deg CW, 1.699m Rev., 32.105deg CCW	Reverse mobility based on MID30 Max.Roll -12.5 deg, max. pitch -9.5 deg (downhill)
30.08.2023 242:19:23:07	032	180deg CW, 0.5m, 180 deg CCW	For Imaging
31.08.2023 243:00:52:00	033	10deg CW, 1m Rev.	Mobility w/o DEM based on visual inspection of Nav Cam image, tot. dist.64m, dis.40m
31.08.2023 243:05:00:00	034	150deg CCW (NavCam), 0.5m, 150deg CW	Tried to take image for westward movement but Sun on NavCam
31.08.2023 243:08:07:30	035	0.5m Rev., 40deg CCW (NavCam)	image taken for movement towards north-east
31.08.2023 243:12:20:00	036	10deg CCW, 4m	Mobility w/o DEM based on visual inspection of Nav Cam image 1m @0.5cm/s for RI + 3m @ 1cm/s
31.08.2023 243:16:39:00	037	4m Rev., 40deg CW (NavCam), 40deg CCW	Decided to retrace due to entry into rim of a crater. This was due to absence of DEM at the end of previous mobility. At -7 deg Roll
01.08.2023 243:21:28:00	038	40deg CW, 1m, 50deg CW, 2m (NavCam), 20deg CCW (NavCam), 20deg CCW (NavCam), 50deg CCW	Mobility w/o DEM based on visual inspection of Nav Cam images
01.09.2023 244:01:20:00	039	70° CW, 2m, 25deg CCW (NavCam), 20deg CCW (NavCam), 25deg CCW (NavCam)	Mobility w/o DEM based on visual inspection of Nav Cam images. Pixel saturation observed in images as a result DEM was not generated. 25 deg CCW obs. as 16 deg in VO
01.09.2023 244:05:40:00	040	35deg CW, 3m, 70deg CCW (NavCam), 20deg CCW (NavCam), 20deg CCW (NavCam), 30deg CW	Due to absence of DEM, entered rim of crater. 3 sets of images taken for panoramic image and DEM
01.09.2023 244:12:55:00	041	20deg CW, 2m (NavCam), 20deg CCW (NavCam)	Mobility w/o DEM based on visual inspection of Nav Cam images. Encountered high variation in roll & pitch & max value of -22 deg & -16

			deg observed in roll & pitch axis respectively. At -11 deg Roll
02.09.2023 244:18:30:31	042	40deg CW, 1m (NavCam,RI), 20deg CCW (NavCam), 20deg CCW (NavCam)	Mobility w/o DEM based on visual inspection of Nav Cam images. At -13.8 deg Roll ATC logic changed to 50deg C to 60degC for W4 sensor as W6 is <0 deg C
02.09.2023 244:22:13:00	043	44.5deg CW, 6.02m (NavCam,RI), 20deg CW (NavCam), 64.5deg CCW	DEM was generated by SAC team. Roll angle computed for the given DEM was coming around 20 deg across 1 st & 2 nd segment. Collective decision taken to go ahead as retrace back would bring to MID 42 origin where DEM generation is not possible. Observed roll of -20 deg for 2-3m traverse
02.09.2023 245:01:47:00	044	50.2deg CW, 5.02m (NavCam), 20deg CW (NavCam), 40deg CCW (NavCam), 30.2deg CCW	Mobility carried out based on DEM. Global slope of -12 deg along roll axis is observed. This path was selected so as to clear the crater rim by next mobility.
02.09.2023 245:05:51:00	045	50.2deg CW, 5m, 50deg CW, 100.2deg CCW	Cleared rim of crater and reached flat terrain at end of mobility. Image taken after 50 deg CW turn to get lander in FOV. This was done to get sense of east direction from current location. Completion of 100 m total distance in this mobility
	046		Commanding issue, uploaded as MID047
02.09.2023 245:08:42:00	047	50.2deg CW, 1m Rev., 50deg CW (NavCam, assuming true East), 80deg CW (NavCam), 20deg CW (NavCam), 10deg CCW (NavCam)	Final mobility End of mission. 50 deg CW turn in 3 rd segment to get Lander in FOV to get sense of east direction from current location. Additional 90 deg turn CW turn taken to park rover facing south.

Appendix-J: Rover NavCam Exposure Settings

PDS Product Id	Exposure Settings 1 (integration lines)	Exposure Settings 1 (integration time ms)	Exposure Settings 2 (integration lines)	Exposure Settings 2 (integration time ms)	Exposure Settings 3 (integration lines)	Exposure Settings 3 (integration time ms)
ch3_nav_ncl_20230823T1801171121_d_img_d32_001	100	35	0	0	320	112
ch3_nav_ncr_20230823T1801572401_d_img_d32_001	100	35	148	51.8	180	63
ch3_nav_ncl_20230824T1415115229_d_img_d32_002	40	14	40	14	40	14
ch3_nav_ncr_20230824T1415423708_d_img_d32_002	40	14	40	14	40	14
ch3_nav_ncl_20230825T1040153807_d_img_d32_003	20	7	20	7	20	7
ch3_nav_ncr_20230825T1040453409_d_img_d32_003	23	8.05	23	8.05	23	8.05
ch3_nav_ncl_20230825T1429025593_d_img_d32_003	30	10.5	30	10.5	30	10.5
ch3_nav_ncr_20230825T1429334084_d_img_d32_003	33	11.55	33	11.55	33	11.55
ch3_nav_ncl_20230825T1951116473_d_img_kru_003	30	10.5	30	10.5	30	10.5
ch3_nav_ncr_20230825T1951424965_d_img_kru_003	33	11.55	33	11.55	33	11.55
ch3_nav_ncl_20230826T1415013468_d_img_d32_004	40	14	40	14	40	14
ch3_nav_ncr_20230826T1415321959_d_img_d32_004	43	15.05	43	15.05	43	15.05
ch3_nav_ncl_20230826T1910175356_d_img_gnh_005	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230826T1910571826_d_img_gnh_005	28	9.8	33	11.55	38	13.3
ch3_nav_ncl_20230827T0344528296_d_img_bl1_006	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230827T0345344591_d_img_bl1_006	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230827T1810173096_d_img_d32_008	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230827T1810585259_d_img_d32_008	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230827T2210172456_d_img_gnh_008	25	8.75	30	10.5	0	0
ch3_nav_ncr_20230827T2210573742_d_img_gnh_008	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230828T0555552932_d_img_d32_010	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230828T0556384939_d_img_d32_010	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230828T1732047656_d_img_d32_013	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230828T1732459183_d_img_d32_013	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230828T2026064296_d_img_gnh_014	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230828T2026475822_d_img_gnh_014	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T0059424569_d_img_kru_013	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T0100242219_d_img_kru_013	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T0528081571_d_img_nno_015	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T0528513579_d_img_nno_015	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T0537587495_d_img_d32_015	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T0538419499_d_img_d32_015	27	9.45	32	11.2	37	12.95

ch3_nav_ncl_20230829T1712052122_d_img_d32_016	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T1712461102_d_img_d32_016	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T1712052122_d_img_d32_021	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T1712461102_d_img_d32_021	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T1915529319_d_img_d32_021	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T1916352939_d_img_d32_021	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T2211041576_d_img_d32_022	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T2211446062_d_img_d32_022	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230829T2326016296_d_img_gnh_025	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230829T2326417583_d_img_gnh_025	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T0201159336_d_img_gnh_024	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230830T0201566228_d_img_gnh_024	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T0411412160_d_img_kru_027	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230830T0412211503_d_img_kru_027	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T0500344616_d_img_nno_026	30	10.5	30	10.5	30	10.5
ch3_nav_ncr_20230830T0501145903_d_img_nno_026	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T1142592296_d_img_nno_029	25	8.75	30	10.5	0	0
ch3_nav_ncr_20230830T1143393582_d_img_nno_029	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T1430486696_d_img_d32_024	896	313.6	896	313.6	896	313.6
ch3_nav_ncr_20230830T1431298859_d_img_d32_024	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T1443301356_d_img_d32_031	30	10.5	30	10.5	30	10.5
ch3_nav_ncr_20230830T1444101423_d_img_d32_031	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T1928548136_d_img_d32_032	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230830T1929359663_d_img_d32_032	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230830T2301162535_d_img_gnh_030	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230830T2301563822_d_img_gnh_030	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T0100412456_d_img_gnh_024	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T0101223983_d_img_gnh_024	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T0100412456_d_img_gnh_031	25	8.75	30	10.5	0	0
ch3_nav_ncr_20230831T0101223983_d_img_gnh_031	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T0505275813_d_img_kru_035	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T0506086702_d_img_kru_035	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T0813019172_d_img_kru_036	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T0813451179_d_img_kru_036	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T1304316922_d_img_nno_037	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T1305142699_d_img_nno_037	27	9.45	32	11.2	37	12.95

ch3_nav_ncl_20230831T1648126376_d_img_d32_038	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T1648537902_d_img_d32_038	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T2136041891_d_img_d32_039	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T2136473900_d_img_d32_039	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T2142009893_d_img_d32_029	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T2142431659_d_img_d32_029	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230831T2147105576_d_img_d32_031	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230831T2147517102_d_img_d32_031	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T0129543336_d_img_gnh_040	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T0130375979_d_img_gnh_040	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T0135213735_d_img_gnh_040	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T0136046379_d_img_gnh_040	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T0213409576_d_img_gnh_040	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T0214218628_d_img_gnh_040	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T0550086695_d_img_gnh_041	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T0550487983_d_img_gnh_041	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T0554005416_d_img_nno_042	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T0554406703_d_img_nno_042	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T1301408296_d_img_d32_042	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T1302209583_d_img_d32_042	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T1312556972_d_img_nno_041	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T1313351983_d_img_nno_041	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T1833314856_d_img_d32_043	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T1834116143_d_img_d32_043	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T1841159976_d_img_d32_034	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T1841561263_d_img_d32_034	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T1846409360_d_img_d32_044	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T1847202223_d_img_d32_044	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T2225595662_d_img_d32_044	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T2226391343_d_img_d32_044	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230901T2233168936_d_img_d32_044	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230901T2233572231_d_img_d32_044	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230902T0157006160_d_img_gnh_045	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230902T0157401903_d_img_gnh_045	27	9.45	32	11.2	37	12.95

ch3_nav_ncl_20230902T0202579496_d_img_gnh_037	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230902T0203387831_d_img_gnh_037	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230902T0202579496_d_img_gnh_045	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230902T0203387831_d_img_gnh_045	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230902T0602577575_d_img_gnh_046	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230902T0603378862_d_img_gnh_046	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230902T0617542696_d_img_gnh_038	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230902T0618343982_d_img_gnh_038	27	9.45	32	11.2	37	12.95
ch3_nav_ncl_20230902T0858238377_d_img_nno_047	25	8.75	30	10.5	35	12.25
ch3_nav_ncr_20230902T0859039663_d_img_nno_047	27	9.45	32	11.2	37	12.95

Appendix-K: Rover NavCam Gain Settings

PDS Product Id	Gain 1	Gain 2	Gain 3
ch3_nav_ncl_20230823T1801171121_d_img_d32_001	2	0	3
ch3_nav_ncr_20230823T1801572401_d_img_d32_001	2	2	2
ch3_nav_ncl_20230824T1415115229_d_img_d32_002	2	2	2
ch3_nav_ncr_20230824T1415423708_d_img_d32_002	2	2	2
ch3_nav_ncl_20230825T1040153807_d_img_d32_003	2	2	2
ch3_nav_ncr_20230825T1040453409_d_img_d32_003	2	2	2
ch3_nav_ncl_20230825T1429025593_d_img_d32_003	2	2	2
ch3_nav_ncr_20230825T1429334084_d_img_d32_003	2	2	2
ch3_nav_ncl_20230825T1951116473_d_img_kru_003	2	2	2
ch3_nav_ncr_20230825T1951424965_d_img_kru_003	2	2	2
ch3_nav_ncl_20230826T1415013468_d_img_d32_004	2	2	2
ch3_nav_ncr_20230826T1415321959_d_img_d32_004	2	2	2
ch3_nav_ncl_20230826T1910175356_d_img_gnh_005	2	2	2
ch3_nav_ncr_20230826T1910571826_d_img_gnh_005	2	2	2
ch3_nav_ncl_20230827T0344528296_d_img_bl1_006	2	2	2
ch3_nav_ncr_20230827T0345344591_d_img_bl1_006	2	2	2
ch3_nav_ncl_20230827T1810173096_d_img_d32_008	2	2	2
ch3_nav_ncr_20230827T1810585259_d_img_d32_008	2	2	2
ch3_nav_ncl_20230827T2210172456_d_img_gnh_008	2	2	0
ch3_nav_ncr_20230827T2210573742_d_img_gnh_008	2	2	2
ch3_nav_ncl_20230828T0555552932_d_img_d32_010	2	2	2
ch3_nav_ncr_20230828T0556384939_d_img_d32_010	2	2	2
ch3_nav_ncl_20230828T1732047656_d_img_d32_013	2	2	2
ch3_nav_ncr_20230828T1732459183_d_img_d32_013	2	2	2
ch3_nav_ncl_20230828T2026064296_d_img_gnh_014	2	2	2
ch3_nav_ncr_20230828T2026475822_d_img_gnh_014	2	2	2
ch3_nav_ncl_20230829T0059424569_d_img_kru_013	2	2	2
ch3_nav_ncr_20230829T0100242219_d_img_kru_013	2	2	2
ch3_nav_ncl_20230829T0528081571_d_img_nno_015	2	2	2
ch3_nav_ncr_20230829T0528513579_d_img_nno_015	2	2	2
ch3_nav_ncl_20230829T0537587495_d_img_d32_015	2	2	2
ch3_nav_ncr_20230829T0538419499_d_img_d32_015	2	2	2
ch3_nav_ncl_20230829T1712052122_d_img_d32_016	2	2	2
ch3_nav_ncr_20230829T1712461102_d_img_d32_016	2	2	2
ch3_nav_ncl_20230829T1712052122_d_img_d32_021	2	2	2
ch3_nav_ncr_20230829T1712461102_d_img_d32_021	2	2	2
ch3_nav_ncl_20230829T1915529319_d_img_d32_021	2	2	2
ch3_nav_ncr_20230829T1916352939_d_img_d32_021	2	2	2
ch3_nav_ncl_20230829T2211041576_d_img_d32_022	2	2	2
ch3_nav_ncr_20230829T2211446062_d_img_d32_022	2	2	2
ch3_nav_ncl_20230829T2326016296_d_img_gnh_025	2	2	2
ch3_nav_ncr_20230829T2326417583_d_img_gnh_025	2	2	2
ch3_nav_ncl_20230830T0201159336_d_img_gnh_024	2	2	2

ch3_nav_ncr_20230830T0201566228_d_img_gnh_024	2	2	2
ch3_nav_ncl_20230830T0411412160_d_img_kru_027	2	2	2
ch3_nav_ncr_20230830T0412211503_d_img_kru_027	2	2	2
ch3_nav_ncl_20230830T0500344616_d_img_nno_026	2	2	2
ch3_nav_ncr_20230830T0501145903_d_img_nno_026	2	2	2
ch3_nav_ncl_20230830T1142592296_d_img_nno_029	2	2	0
ch3_nav_ncr_20230830T1143393582_d_img_nno_029	2	2	2
ch3_nav_ncl_20230830T1430486696_d_img_d32_024	1	3	3
ch3_nav_ncr_20230830T1431298859_d_img_d32_024	2	2	2
ch3_nav_ncl_20230830T1443301356_d_img_d32_031	2	2	2
ch3_nav_ncr_20230830T1444101423_d_img_d32_031	2	2	2
ch3_nav_ncl_20230830T1928548136_d_img_d32_032	2	2	2
ch3_nav_ncr_20230830T1929359663_d_img_d32_032	2	2	2
ch3_nav_ncl_20230830T2301162535_d_img_gnh_030	2	2	2
ch3_nav_ncr_20230830T2301563822_d_img_gnh_030	2	2	2
ch3_nav_ncl_20230831T0100412456_d_img_gnh_024	2	2	2
ch3_nav_ncr_20230831T0101223983_d_img_gnh_024	2	2	2
ch3_nav_ncl_20230831T0100412456_d_img_gnh_031	2	2	0
ch3_nav_ncr_20230831T0101223983_d_img_gnh_031	2	2	2
ch3_nav_ncl_20230831T0505275813_d_img_kru_035	2	2	2
ch3_nav_ncr_20230831T0506086702_d_img_kru_035	2	2	2
ch3_nav_ncl_20230831T0813019172_d_img_kru_036	2	2	2
ch3_nav_ncr_20230831T0813451179_d_img_kru_036	2	2	2
ch3_nav_ncl_20230831T1304316922_d_img_nno_037	2	2	2
ch3_nav_ncr_20230831T1305142699_d_img_nno_037	2	2	2
ch3_nav_ncl_20230831T1648126376_d_img_d32_038	2	2	2
ch3_nav_ncr_20230831T1648537902_d_img_d32_038	2	2	2
ch3_nav_ncl_20230831T2136041891_d_img_d32_039	2	2	2
ch3_nav_ncr_20230831T2136473900_d_img_d32_039	2	2	2
ch3_nav_ncl_20230831T2142009893_d_img_d32_029	2	2	2
ch3_nav_ncr_20230831T2142431659_d_img_d32_029	2	2	2
ch3_nav_ncl_20230831T2147105576_d_img_d32_031	2	2	2
ch3_nav_ncr_20230831T2147517102_d_img_d32_031	2	2	2
ch3_nav_ncl_20230901T0129543336_d_img_gnh_040	2	2	2
ch3_nav_ncr_20230901T0130375979_d_img_gnh_040	2	2	2
ch3_nav_ncl_20230901T0135213735_d_img_gnh_040	2	2	2
ch3_nav_ncr_20230901T0136046379_d_img_gnh_040	2	2	2
ch3_nav_ncl_20230901T0213409576_d_img_gnh_040	2	2	2
ch3_nav_ncr_20230901T0214218628_d_img_gnh_040	2	2	2
ch3_nav_ncl_20230901T0550086695_d_img_gnh_041	2	2	2
ch3_nav_ncr_20230901T0550487983_d_img_gnh_041	2	2	2
ch3_nav_ncl_20230901T0554005416_d_img_nno_042	2	2	2
ch3_nav_ncr_20230901T0554406703_d_img_nno_042	2	2	2
ch3_nav_ncl_20230901T1301408296_d_img_d32_042	2	2	2
ch3_nav_ncr_20230901T1302209583_d_img_d32_042	2	2	2
ch3_nav_ncl_20230901T1312556972_d_img_nno_041	2	2	2

ch3_nav_ncr_20230901T1313351983_d_img_nno_041	2	2	2
ch3_nav_ncl_20230901T1833314856_d_img_d32_043	2	2	2
ch3_nav_ncr_20230901T1834116143_d_img_d32_043	2	2	2
ch3_nav_ncl_20230901T1841159976_d_img_d32_034	2	2	2
ch3_nav_ncr_20230901T1841561263_d_img_d32_034	2	2	2
ch3_nav_ncl_20230901T1846409360_d_img_d32_044	2	2	2
ch3_nav_ncr_20230901T1847202223_d_img_d32_044	2	2	2
ch3_nav_ncl_20230901T2225595662_d_img_d32_044	2	2	2
ch3_nav_ncr_20230901T2226391343_d_img_d32_044	2	2	2
ch3_nav_ncl_20230901T2233168936_d_img_d32_044	2	2	2
ch3_nav_ncr_20230901T2233572231_d_img_d32_044	2	2	2
ch3_nav_ncl_20230902T0157006160_d_img_gnh_045	2	2	2
ch3_nav_ncr_20230902T0157401903_d_img_gnh_045	2	2	2
ch3_nav_ncl_20230902T0202579496_d_img_gnh_037	2	2	2
ch3_nav_ncr_20230902T0203387831_d_img_gnh_037	2	2	2
ch3_nav_ncl_20230902T0202579496_d_img_gnh_045	2	2	2
ch3_nav_ncr_20230902T0203387831_d_img_gnh_045	2	2	2
ch3_nav_ncl_20230902T0602577575_d_img_gnh_046	2	2	2
ch3_nav_ncr_20230902T0603378862_d_img_gnh_046	2	2	2
ch3_nav_ncl_20230902T0617542696_d_img_gnh_038	2	2	2
ch3_nav_ncr_20230902T0618343982_d_img_gnh_038	2	2	2
ch3_nav_ncl_20230902T0858238377_d_img_nno_047	2	2	2
ch3_nav_ncr_20230902T0859039663_d_img_nno_047	2	2	2